The table of contents

The abstract 2

The introduction 3

The materials and methods 4

The materials 4

The methods **6**

Test Plan: 7

Results: 7

The analysis 9

The conclusion 9

Recommendations 10

The literature cited 12

The acknowledgement 13

The Abstract

Egypt encounters multiple challenges that hinder its growth towards self-independence. The project aims to solve the widespread e-waste crisis, promote recycling and sustainability, and provide an affordable alternative for school labs. This will be done through solving Egypt's grand challenges: improving the scientific and technological environment for all, reducing and adapting to the effects of climate change, recycling garbage and waste for economic and environmental purposes, and addressing and reducing pollution fouling our air, water, and soil. The purpose of the study is to utilize the abundant, unused electronic waste by turning it into a fully-functional bacteriological incubator capable of serving in school labs. The selected problem to be solved is to increase the capabilities of research in Egypt's laboratories through recycling electronic waste. The device was constructed by recycling electronics from old devices, such as a PTC heater and a TEC cooler. Various tests have been done to ensure that the calculation error percentage does not exceed ±5%. There were some negative results overcome by different modifications. Major findings were the device's ability to generate temperatures ranging from 27 °C and 45 °C under average usage circumstances, with the ability to go beyond this range if necessary, while major conclusions ensured that the prototype solved the problem.

The Introduction

Egypt faces multiple grand challenges that undeniably slow its development. One of these is the increasing amount of unrecycled e-waste in the environment. As of 2022, the amount of e-waste generated in Egypt is estimated to be ~690,000 tons, which is 17.76% higher than the ~586,000 tons generated in 2019. While reports cite roughly 150,000 collectors + ~50,000 waste processors active in Egypt, the documented formal collection is extremely low in some national studies, as one assessment reported only ~1,584 tons of e-waste is being collected every year, which is only around 2-3% of the yearly generated amount. In addition, only around 20% may undergo local recycling, while the remainder is sent to uncontrolled landfills or dumpsites or stockpiled. All these amounts of unrecycled e-waste result in high heavy-metal levels (Pb, Zn, Cu, Mn) in the soil, more specifically in industrial zones. Some studies measured Pb values to reach as high as ~470 mg/kg, and around a major lead smelter site in Egypt, researchers measured extremely elevated Pb in nearby soils (example peak ~3,130 mg/kg).

Therefore, our selected problem to be solved is to increase the capabilities of research in Egypt's laboratories through recycling electronic waste. In the meantime, e-waste is one of the severe crises that threatens the residents who live near industrial zones. Hence, solving this problem would successfully reduce the amount of e-waste and heavy metals in the environment, in addition to turning it into a useful project that can be used in school laboratories.

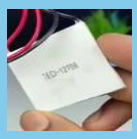
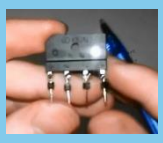
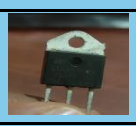
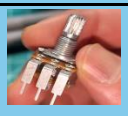
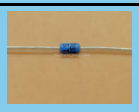
To gather as much beneficial data for the project as possible, different previous solutions for the problem were reviewed. The most outstanding of which were the VEVOR 20L lab incubator and the Thermo Scientific

Heratherm IGS180. The common points of strength present in both were the ability to generate temperatures where most types of bacteria can survive and grow ordinarily, and the ability to operate silently.

After reviewing the prior solutions, the selected solution is to construct a lightweight bacterial incubator using unrecycled electronic waste. The incubator will meet a design requirement criterion of performing at least one laboratory application, in addition to having two extra functions.

The Materials and Methods

The materials

Picture							
Name	PTC Aluminum Heater Plate	Electric Transformers 12V & 16 V	Thermoelectric Cooler	RECTIFIER BRIDGE	Home AC unit Thermostat	PC Heatsink and Fan	Old Oven Frame
Quantity	2 pieces	2 pieces (16 V 10 A, and 12 V 1 A)	1 piece	2 pieces	1 piece	1 piece	1 piece
Description	a heating element of the convection system of the prototype	Turns the unstable alternating current found in households into usable DC.	A cooling element of the convection system of the prototype	Turns the Alternating (AC) current into direct current (DC)	To control the temperature of the prototype	Used for heating and cooling in the convection system of the prototype	The frame of the prototype
Picture							
Name	Recycled acrylic	Triac bt136	Capacitors	Potentiometer	Wires	silicon diodes	Diac db3

Quantity	1 Pieces	1 piece	2 pieces (4500 micro 50 volts, and 470 micro 25 volts)	2 pieces	From 1 to 1.5 m	2 pieces	1 piece
Description	To make the side of the frame	Used in the dimmer circuit to control the light intensity	Softens Direct Current (DC) coming out of the rectifier bridges	Two potentiometers one with 5 kilo ohms and other with 1 mega ohm	A copper wire to connect the circuit	Used for setting the speed of the fan for the heating and the cooling system separately	Used in the dimmer circuit to control the light intensity

The methods

1. For the sake of constructing a successful prototype, a 3D design was prepared first to visualize the prototype's shape, **which is shown in Figure 1.**
2. Disassembled the oven frame, exposed the fiber glass, extracted the fiber glass insulation layer from the oven, and Extracted the PTC heaters and rotary switches of the oven.
3. Removed non-essential components remaining in the oven, cleaned the empty frame of the oven of left over residue **as shown in Figure 2.**
4. Extracted the power supply from the PC to extract the capacitor, wires and the variable resistance.
5. Extracted the rectifier bridge, a capacitor, and the acrylic from the mini fridge.

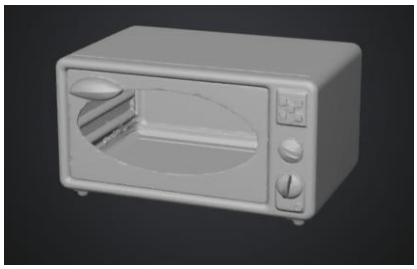


Figure 1 3D Design



Figure 2 Prototype Residue Extraction

6. The Thermostat was extracted from a home AC unit, and the transformer 16V 10A was extracted from the oven while the 12V 1A transformer was extracted from an old radio.
7. Connected the transformer with the capacitor and the bridge to soften the direct current, put the heat sink in the oven, and the fan.



Figure 3 Heat Plate Placement

8. Put the heater plates on the side of heat sink, **as shown in Figure 3**, and the thermoelectric on the back of the oven. Connected the thermostat to the prototype panel and connected silicon diodes on the PTC and thermoelectric cooler wires to the fan to adjust fan speed using variable resistance.
9. Soldered a dimmer circuit from a capacitor, triac, diac, resistor, and variable resistance to control the light intensity, and connected it to an old potentiometer cap from the oven.
10. Established a joint connection between one side of all components, connected the remaining sides to an old rotary switch from the oven.
11. Connected the PTC and thermoelectric wires to the relay of the thermostat.
12. Connected the heat circuit to the transformer output.
13. Completed the final connections of the circuit as shown in the schematic **in Figure 4.**

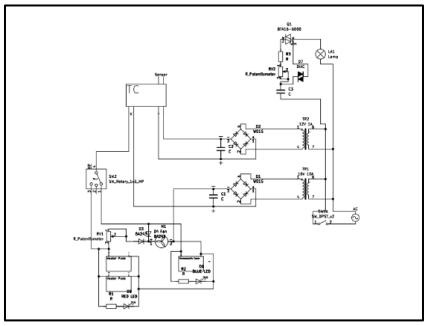


Figure 4 MultiCubator Circuit Diagram

Test Plan:

A test plan ensuring the solution is both working properly and meeting the design requirements consisted of the following:

1. Using a voltameter, the voltage of the electric current at the point of the PTC heaters and the TEC cooler is measured, ensuring the functionality of the transformer.
2. The temperature in the incubator was measured by putting a thermometer inside it for 5 minutes, ensuring the calibration validity of the heat controller thermostat.
3. A digital thermometer is installed on the top of the incubator's frame, ensuring the quality of the insulation.
4. An E. coli bacteria sample is prepared and placed inside the incubator for 6 hours, testing the growth rate.

Results:

Negative results:

- While disassembling the conventional oven, the frame almost broke due to trying to unscrew it from the wrong side. It was then reassembled, and the legs were properly removed first.
- When the voltage was measured, it was found that the transformer's capacity was not enough for the PTC requirements, so multiple other transistors were used until one worked.

- During the calibration test, it was found that the calibration was off by 0.9°C, then 0.4°C. the thermostat was recalibrated until it matched the real temperature.

Positive results:

- The voltage of the current was exactly 12V, matching the PTC heaters.
- The thermostat calibration matched the thermometer's readings perfectly.
- The digital thermostat measured no change in temperature outside of the incubator, indicating that the insulation is working.
- The measured growth of the E.coli bacteria sample showed exponential growth, **which is presented in table 1 and Figure 5**.

Table 1: Cell/Colony Growth

Time (h)	Cells/colony
0	1
1	8±4
2	64±8
3	512±16
4	4,096±32
5	32,768±64
6	262,144±127

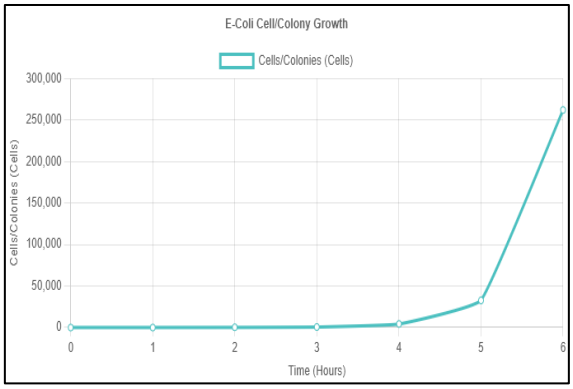


Figure 5 E-Coli Cell/Colony Growth

The Analysis

The Conclusion

The problem chosen to be solved for this project was to turn electronic waste into a beneficial device able to serve in school laboratories. The selected solution was to construct a bacteriological incubator, as it has the potential to utilize multiple different wasted electronic parts and make a real difference in the school's biology lab. The device was made of a heating system, a cooling system, and an old oven functioning as the shell of the prototype. Regardless of the negative results, the prototype has successfully met all the major design requirements of performing at least two functions (generating heat and controlling light intensity), and providing at least one measurement (the internal chamber's temperature). Moreover, the device was able to overcome the high initial price problem present in the two previous solutions, the VEVOR 20L lab incubator and the Thermo Scientific Heratherm IGS180. By reviewing and benefiting from other solutions, the device has successfully increased the research capabilities in school labs, besides turning e-waste into something beneficial.

Recommendations

1.Humidity control system

It is highly recommended to add a humidity control system to the device due to its ability to measure humidity levels in the chamber and make adjustments as needed. Modifying the incubator to utilize this system would enable it to preserve the agar from evaporating for a longer duration, allowing for longer research periods without needing to replace the agar plates frequently. However, implementing this suggestion is challenging due to safety concerns and material scarcity. Humidity sensors are extremely rare to find in older devices, and even if they can be acquired, they pose potential safety risks. Humidity control systems require a strong sealing mechanism to minimize moisture exposure near heating elements and electronics. Otherwise, water could damage the circuitry and potentially pose a hazard to users. A final concern to address is accuracy. Constructing a humidity system using scavenged materials while maintaining an error percentage within ±5% would be an extremely challenging task.

CO2 injection system:

An important addition to be recommended for the incubator is a CO2 injection system. This is because most of the culture media contain sodium bicarbonate, which is a base. The CO2 reacts with the water, which forms carbonic acid. This neutralizes the media. Otherwise, the system would lose

acidity, and the cells would eventually die in the long term. The CO2 injection system consists of an IR sensor that detects the CO2 levels in the incubator, and when it drops below 5%, it opens a solenoid valve that's connected to a CO2 gas cylinder, puffing the gas into the incubator. **This system is presented in Figure 6.** We unfortunately couldn't apply such a system in our prototype, since, without a license, we are not allowed to have a CO2 container for safety reasons.

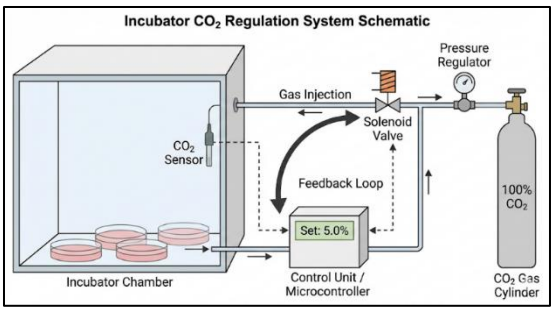


Figure 6 CO2 Injection System

Water jacket insulation system:

The most important section in the incubator is the insulation system, since poor insulation would lead to huge amounts of heat losses, which would put the heaters under a higher load just to reach the desired temperature. For instance, if the heaters worked regularly, they would need to be active every 3 minutes; poor insulation would drastically increase that to once every minute. This also has a higher risk of killing the cells, since if the walls get too cold, water droplets will start to build up on the walls and the ceiling, and if a droplet falls on the petri dish, this instantly contaminates and kills the cells. We can use the water jacket insulation system to insulate the incubator, which works by double-layering the walls and pouring warm water in between. Water has high heat capacity, which is why it's a good insulator, and it also matches our requirement of the project being environmentally friendly. We instead used passive foam, since the water jacket requires stainless steel, which is hard to find in old devices.

Stainless steel & copper lining:

It's recommended to double-layer the body of the incubator, where it's best to use stainless steel for the outer shell, which has high chemical resistance, preventing rusting. However, for the inner chamber and racks, it's recommended to use copper lining as the main material, **as shown in figure 7.** Biofilm has a high risk of cross-contamination between the bacteria, where the bacteria produce an extracellular matrix that sticks to surfaces, and rough surfaces like wood or plastic are easier to stick to. To prevent biofilm, the best way is to use copper lining. This totally prevents any leakage of bacteria outside the petri dish. That's because copper lining is oligodynamic, which means that it instantly kills any bacteria that touch it. This is crucial, since the one thing that shouldn't happen in a laboratory is contamination in/outside the incubators, which can cause catastrophic disease spreads. This wasn't applied in the prototype because copper lining costs too much, reaching 11 USD per kg.

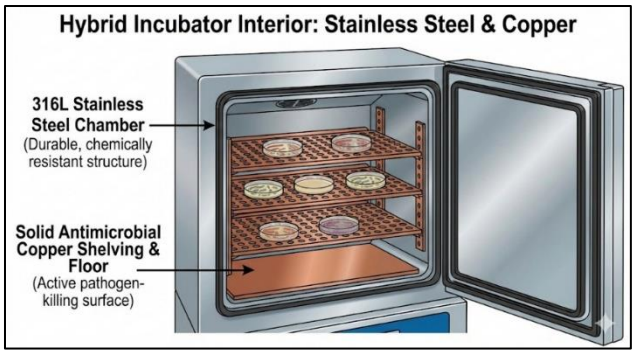


Figure 7 Steel & Copper Lining

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